

HAND IN

answers record on exam paper

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Queens University Final Examination
Smith Engineering
Department of Electrical And Computer Engineering

ELEC 477 Distributed Systems

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April 22, 2025

Student Number _____

Student Number in words _____
(Example: two three two nine seven four two five)

INSTRUCTIONS TO STUDENTS:

- This examination is 3 HOURS in length.
- Answer all questions in the space provided on these pages. If you need extra space, use the extra pages at the end of the exam.
- **No aids are allowed.**
- This exam has 8 pages
- Total 36 marks. Please pay attention to the number of marks for each question.
- The numbers in the bottom right of the page show a running total of the marks in the exam (8/30 on page 5 means 8 marks on page 5 and 30 marks for all pages up to and including page 5)

PLEASE NOTE:

Proctors are unable to respond to queries about the interpretation of exam questions.
Do your best to answer exam questions as written.

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1. (4 marks) Give a brief (one or two line) description of each of the following terms as they were discussed in class.

a. Deliberate Failure

b. Utilization Factor

c. Event Driven

f. Network Time Protocol

2. (2 marks) In what situation would you set a timeout on a socket when implementing a Remote Procedure Call (RPC)?

2. **(2 marks)** Data consistency models are independent of any implementation of a data consistency protocol. How do data consistency models help in the implementation of a distributed system?

5. **(2 marks)** In real time systems, there are two different distributions used to characterize the real time behaviour of a system. They are variation in the ____ of a task and variation in the ____ of a result. Identify and describe these two distributions.

4. **(2 marks)** Distributed mutual exclusion is used to ensure only one node has access to a shared resource. Two solutions were a central coordinator and decentralized voting. For a given application how you would decide between them?

5. **(2 marks)** There are two ways of updating the value of a replica, by *copying the update operation* or by copying data. What characteristics of the data would guide your choice of the two operations ?
6. **(2 marks)** Identify and describe the two main reasons for replication in distributed systems.
7. **(2 marks)** We have covered a variety of election algorithms from bullying to distributed voting with timeouts. Some add information that restricts whether a node can vote for another node. What, in general, is the reason to restrict the votes that can be cast by a node?

8. (4 marks) Consider the three periodic processes to be scheduled on a single CPU:

τ_i	C_i	T_i
1	1	8
2	2	5
3	2	9

Provide a schedule for these process for the first 18 time units using the The Earliest Deadline First (EDF) scheduling algorithm. Identify any periods of time where none of the processes are running (i.e. the CPU is idle). Use a vertical bar indicate a boundary between two processes or a boundary between a process and an idle period.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

9.(4 marks). Name servers are used to assign names to entities. Two approaches to namespaces are flat and structured name spaces. What are the pros and cons of each, and how would you choose between them.

10. (5 marks) The raft algorithm makes 5 guarantees, listed below. Provide a definition for each.

a. Election Safety

b. Leader Append Only

c. Log Matching

d. Leader Completeness

e. State Machine Safety

12. (5 marks) Answer true or false by circling the correct answer.

In DDS, there is one queue per instance when the topic has a key. TRUE FALSE

When the Mars Pathfinder first landed, the bus was regularly reset because of a priority inversion. TRUE FALSE

A remote procedure call client stub must be statically configured to know the name and port of the RPC server. TRUE FALSE

In rate monotonic scheduling, process with longer periods get higher priority TRUE FALSE

In a renewal frequency lease, if a replica pulls each time the lease expires, then the lease isn't long enough. TRUE FALSE

Extra Space.

Extra Space.